UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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| COURSE | COMPUTER ORGANIZATION AND ARCHITECTURE |
| COURSE CODE | CSC429 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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**PART A (80 MARKS)**

Answer ALL questions in this part.

**QUESTION 1** (5 marks)

John Von Neumann is a mathematician who is one of the pioneers in computer advancement, and his contributions are still relevant to the modern computer.

a) Draw the Von Neuman architecture and give relevant examples each.

b) What is the name of the actual device that John Von Neumann himself developed?

**QUESTION 2** (5 marks)

Artificial intelligence (A1) is the computer ability to exhibit human intelligent behaviors such as smart home applications, robots, intelligent tutoring systems and others.

a) In which generation is A1 introduced?

b) What is the programming language used for this technology?

c) What is the advantage of embedding A1 in our everyday life?

d) A1 can also posed as a threat to community. Discuss ONE (1) possible scenario when A1 can harm people.

**QUESTION 3** (5 marks)

Solve the following equations in binary number bases. Show your workflow.

a) 1.0000 10012-0.0011 OOH2

b) O.FA3i6-0.651s

**QUESTION 4** (5 marks)

Given the following Figure 1.

a) Label the addressing technique illustrated in Figure 1.

b) Explain how the addressing technique in 9(a) is used.

c) List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in 9(a).

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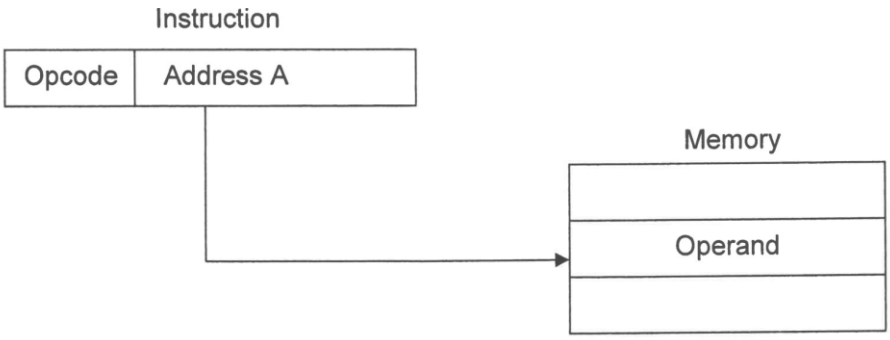



*Figure 1*

**QUESTION 5** (10 marks)

Keyboard is the most common means for computer interaction.

a) Explain TWO (2) types of characters that keyboard provides.

b) Describe THREE (3) categories of input and output module. Give an appropriate example for each category.

**QUESTION 6** (10 marks)

a) Draw Single Instruction, Single data (SISD) architecture with the following information:
- CU is the control unit
- PU is the processing unit
- MU is the memory unit
- IS is the instruction stream
- DS is the data stream

b) List TWO (2) examples of shared memory Multiple Instruction Multiple Data (MIMD).

c) Discuss the difference between what happens if a virus attacks shared memory Multiple Instruction Multiple Data (MIMD) and distributed memory Multiple Instruction Multiple Data (MIMD).

d) Explain ONE (1) benefit of using distributed memory Multiple Instruction, Multiple Data stream (MIMD) architecture with example.

**QUESTION 7** (10 marks)

a) List FOUR (4) classification of multiple processor architecture.

b) Symetric Multi Processor (SMP), Non Uniform Memory Access (NUMA) and Cluster are part of the X multiple processor architecture. Identify X?

c) Differentiate the multiple processor architecture for SMP and Cluster. In your opinion, which architecture is easier to manage and control by giving appropriate justification.

**QUESTION 8** (10 marks)

a) Convert the following values to the respective number bases. Show the complete work flow.

i. 1210.331s to hexadecimal

ii. 295.8125io to octal

iii. BACAieto binary

**QUESTION 9** (10 marks)

System interconnection is a mechanism that provides communication among computer components.

a) What is the size of a bus if the word size is 64 bit?

b) List the TWO (2) types of bus.

c) Illustrate the location of a bus between CPU and memory.

d) Differentiate between system interconnection and CPU interconnection.

e) Discuss ONE (1) drawback each for the different types of bus timing.

**QUESTION 10** (10 marks)

Processor contains a number of registers used for temporary storage of data. Describe FIVE (5) types of registers.

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**PART B (20 MARKS)**

Answer the question in this part.

**QUESTION 11** (20 marks)

Question 1 is based on Figure 2 and Table 1.
write an appropriate program using correct assembly language notations to compute the equation snbwn in Figure 2 using instructions for zero-address, one-address, and three-address. The available instructioni are provided in Tabte 1. The register labeled A, B, C, and D are fixed, white registers x, y, and z can be used temporarily. The final answer must be stored in register z.

$$z = A^2 - B * C + \frac{2}{D + 5}$$

*Figure 2*

*Table 1: Machine instruction*

| Zero-address | One-address | Three-address |
|---|---|---|
| PUSH p | LOAD p | ADD (p,q,r) |
| POP p | STOR p | suB (p,q,r) |
| ADD | ADD p | MUL (p,q,r) |
| SUB | SUB p | DIV (p,q,r) |
| MUL | MUL p | |
| DIV | DIV p | |